MARINDUQUE STATE UNIVERSITY
COLLEGE OF ENGINEERING
DEPARTMENT OF COMPUTER ENGINEERING
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CPPC 21/L
DIGITAL SIGNAL PROCESSING

ACTIVITY No. 1

Submitted by:

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BSCpE 3B

Submitted to:

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Subject Professor

Activity 1: Numerical Methods + Wave Animation (Heun's, Implicit Euler, RK4)

Objectives

- Solve differential equations numerically
- Compare accuracy of different methods
- Visualize wave motion

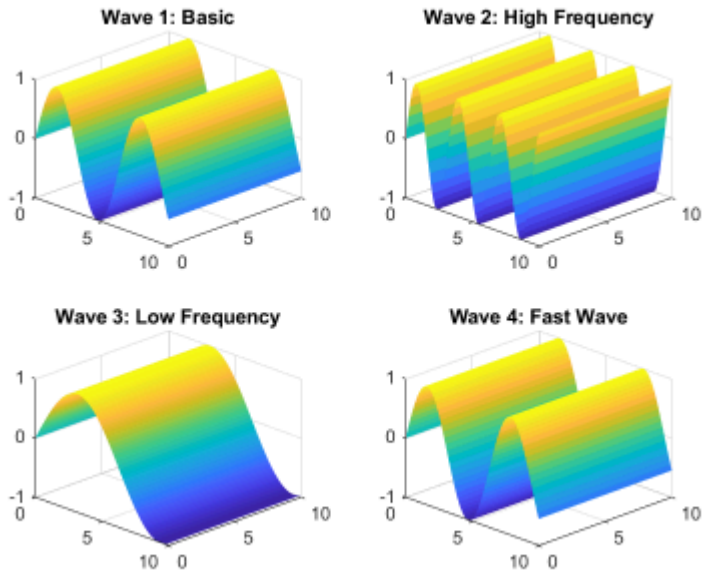
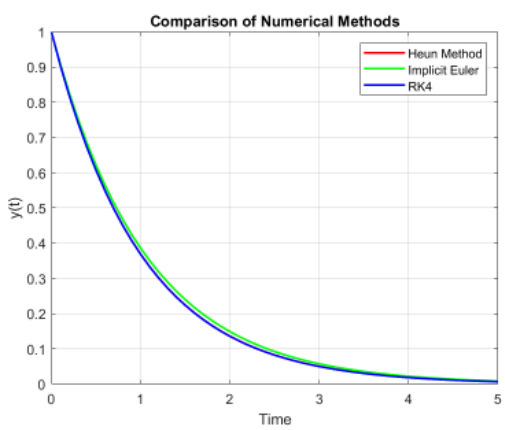
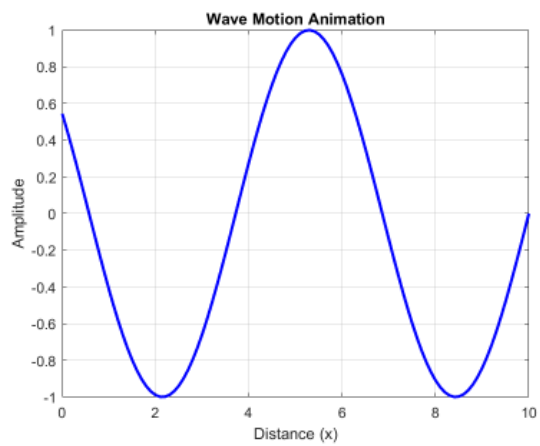
Materials

- MATLAB software

Procedures

1. Define differential equation
2. Implement:
    - Heun's Method
    - Implicit Euler Method
    - Runge-Kutta 4th Order
3. Compare results
4. Create wave animation using `plot` and `pause`

Conclusion

Higher-order methods like RK4 provide more accurate results compared to simpler methods.

Documentations